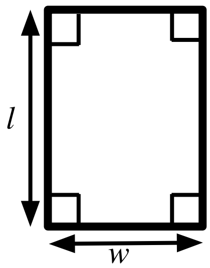
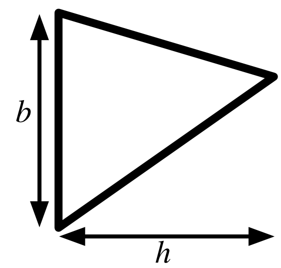
Problem solving with expressions and formulae

Task A: Forming expressions for area

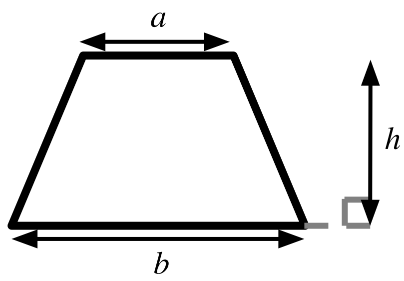
Here are some formula for the area of 2D shapes which may help you with the following task.



area = width × length



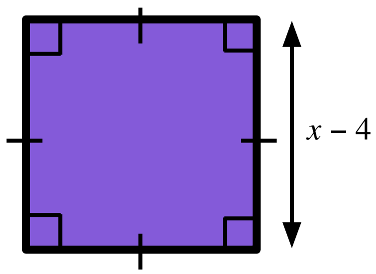
area = $\frac{1}{2}$ × base × height



area = $\frac{1}{2} \times (a + b) \times$ height

Where a and b are the parallel sides.

1) Write a sentence explaining why the area of this shape can be written in each of the following ways:



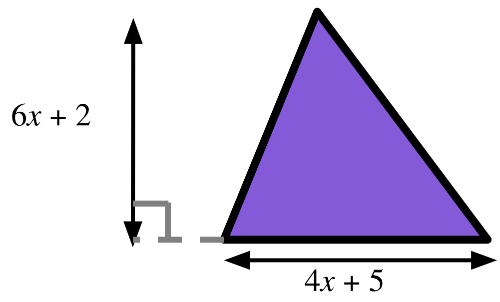
a) $(x - 4)^2$

b) $(x - 4)(x - 4)$

c) $x^2 - 8x + 16$

2) Draw a rectangle with an area of $(6 + y)(2 + y)$ write a different expression for the area of your rectangle.

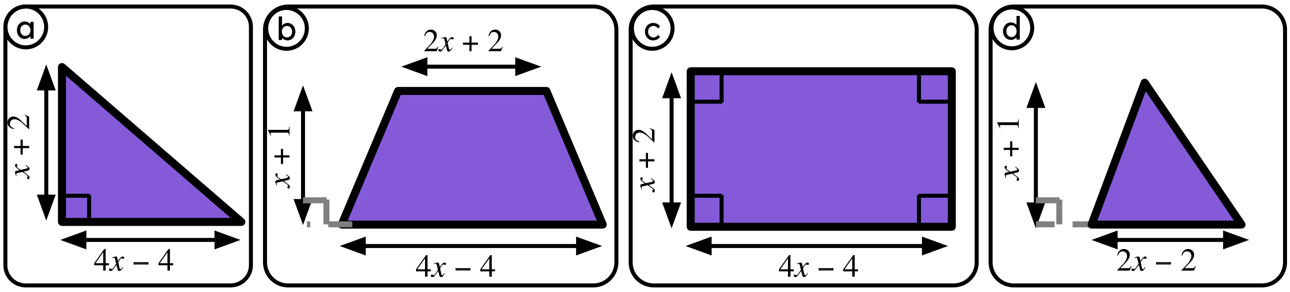
3a) Explain why the area of this shape be written as $\frac{1}{2}(4x + 5)(6x + 2)$



b) Write the expression for the area of this shape in two different ways.

c) Draw and label a triangle with area $(x + 1)(3x - 2)$

4) Match these shapes to the correct expression for their area.



e) $x^2 - 1$

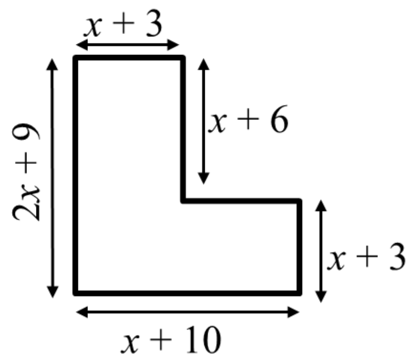
f) $2x^2 + 2x - 4$

g) $3x^2 + 2x - 1$

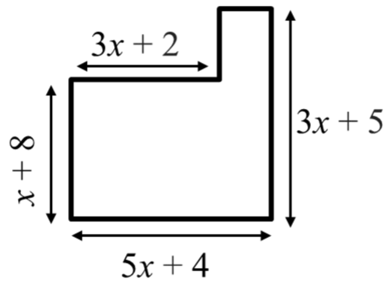
h) $4x^2 + 4x - 8$

Task B: Area of compound shapes

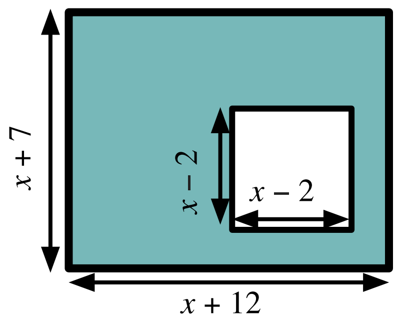
1) Calculate a simplified expression for the area of this shape.



2) Calculate a simplified expression for the area of this shape.

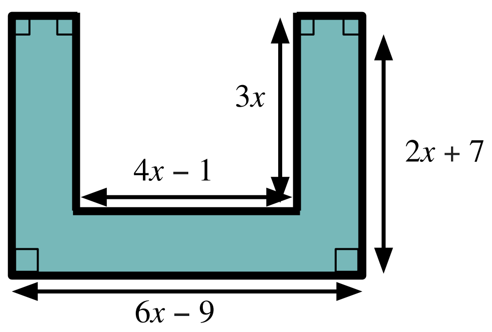


3a) Calculate a simplified expression for the area of the shaded shape.



b) The area of the shaded shape is 149 square units. Use this to find the value of x

4a) Calculate a simplified expression for the area of the shaded shape.

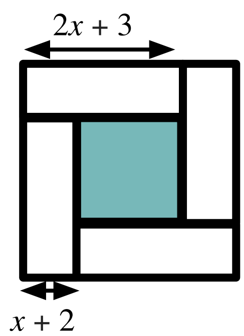


Not drawn to scale

b) The area of the shaded shape is 99 square units. Use this to find the value of x

5) Four identical rectangles sit round the outside of a shaded square.

a) Find a simplified expression for the area of one rectangle.



b) Find a simplified expression for the area of the whole shape.

c) Find a simplified expression for the area of the shaded square.